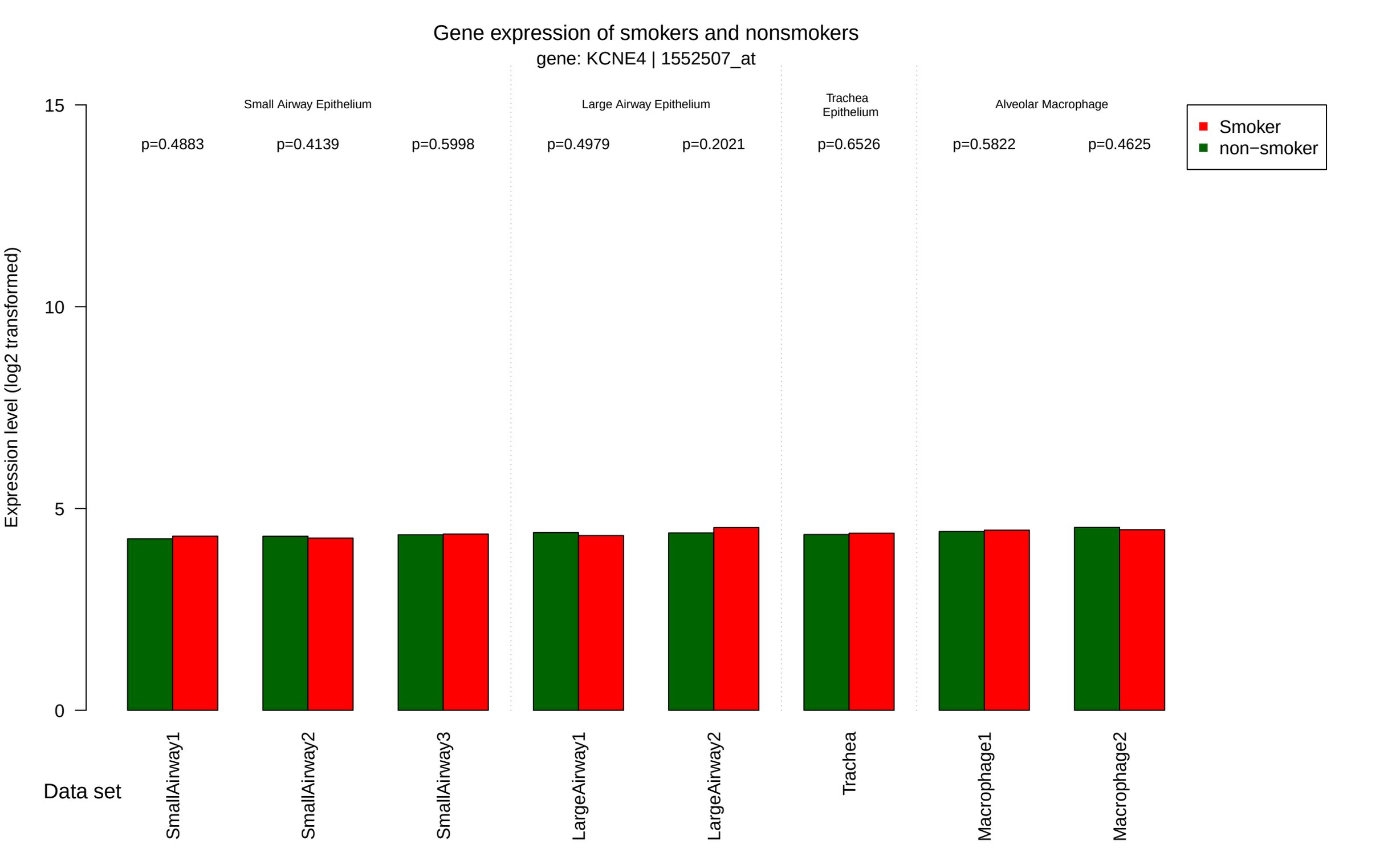
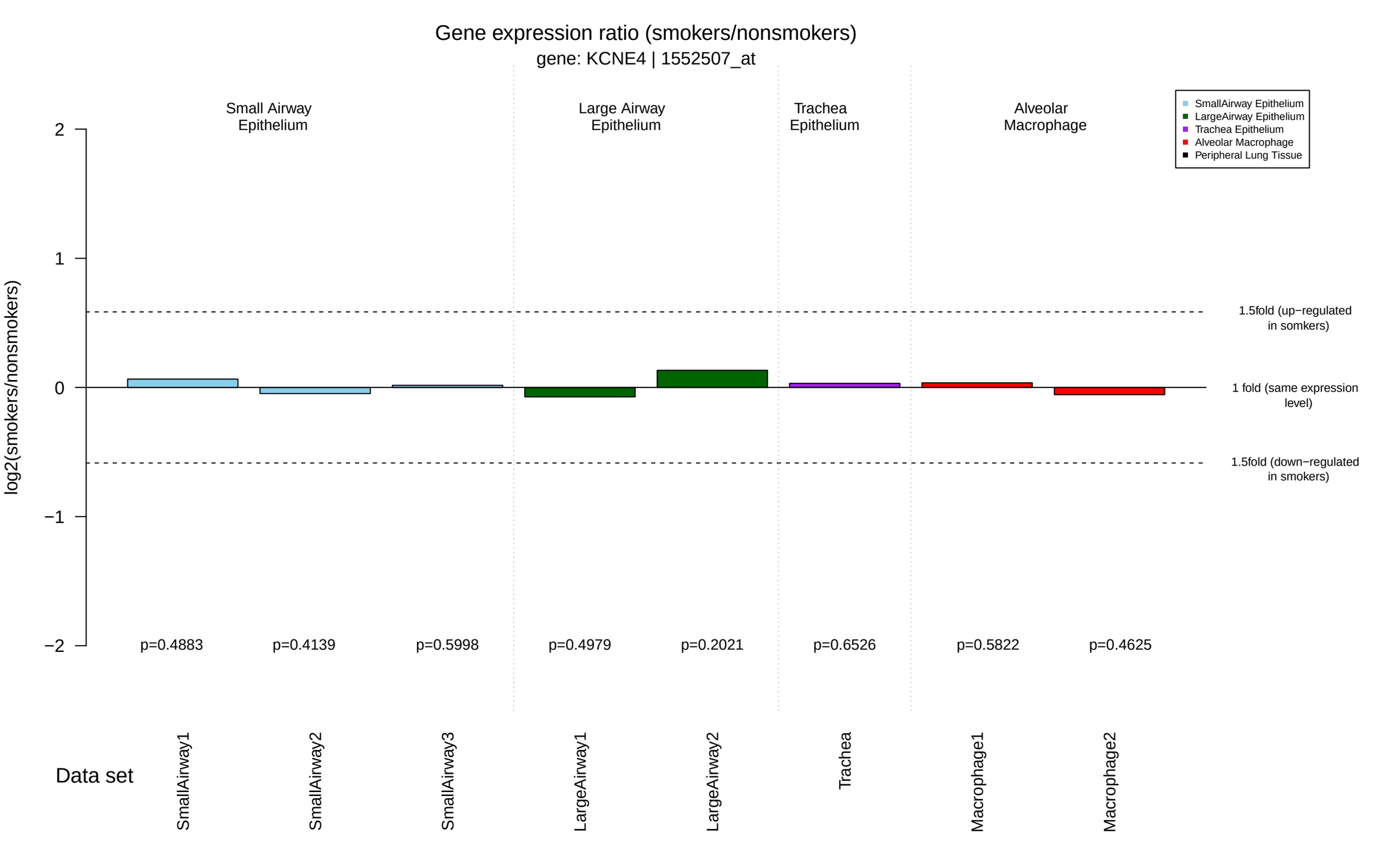
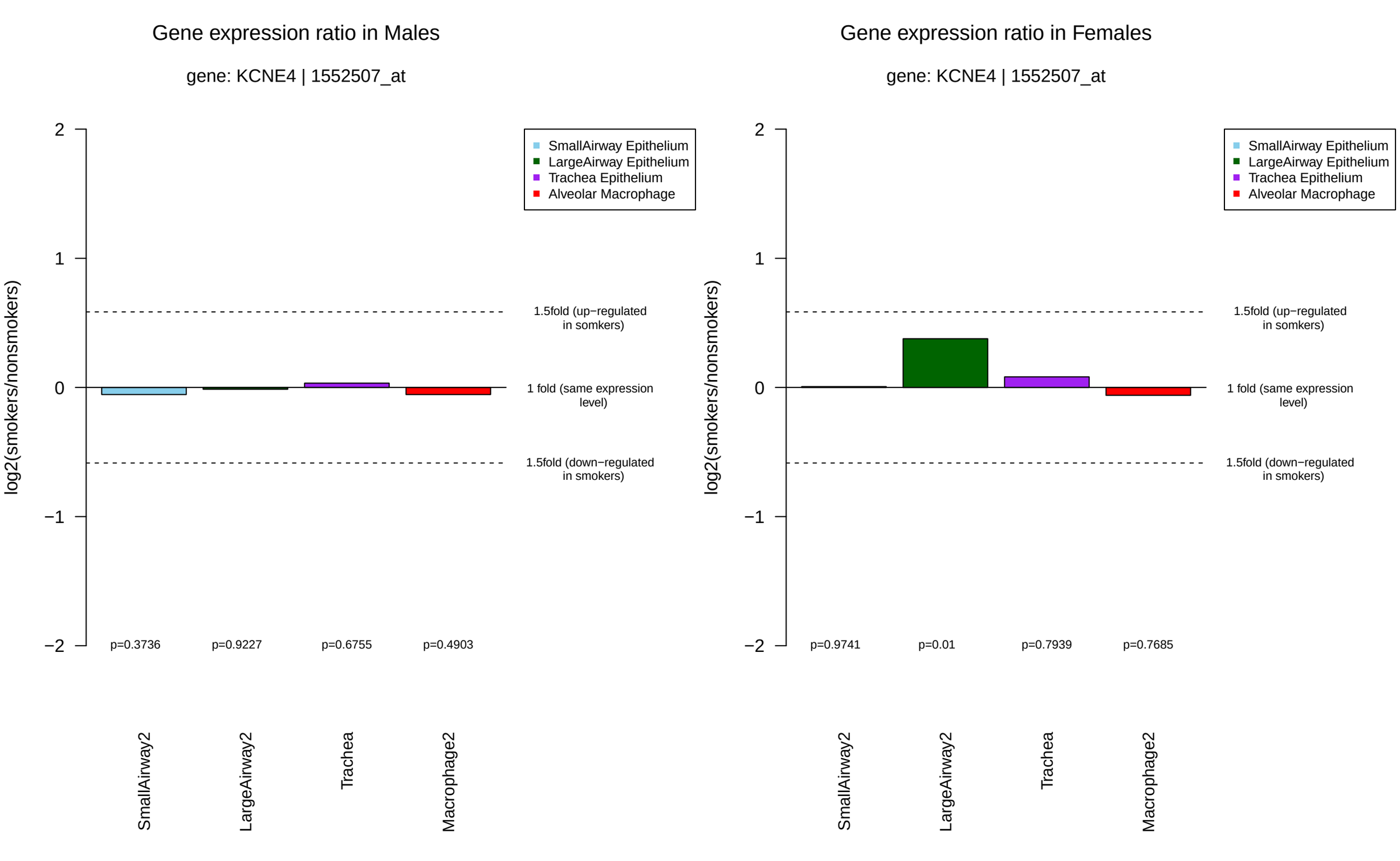
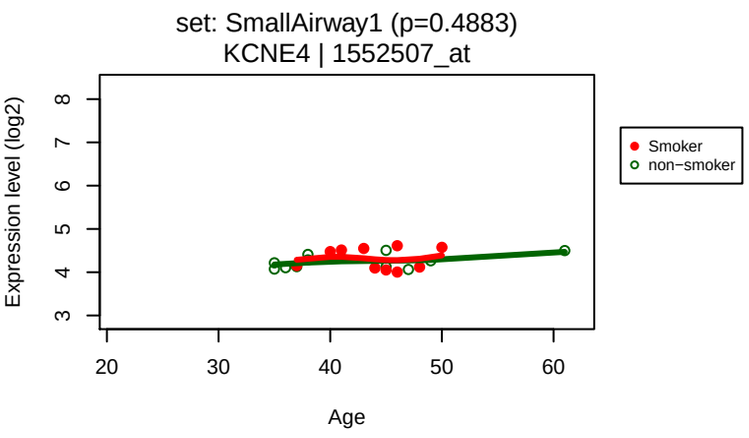
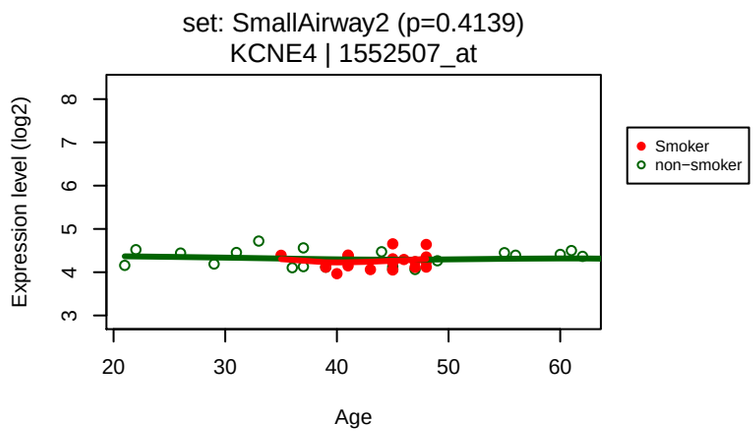
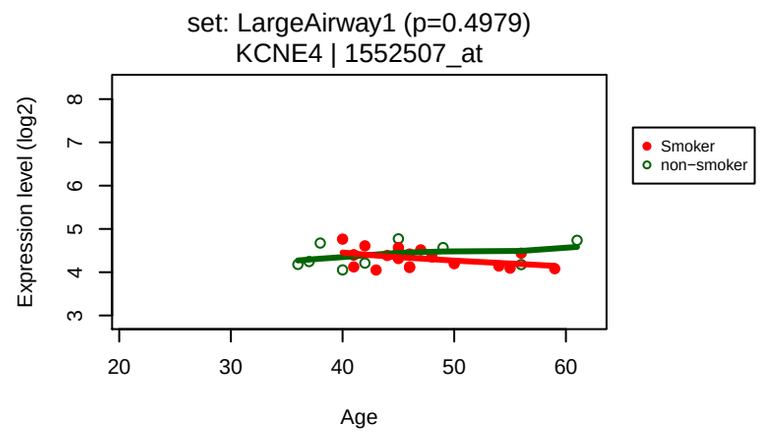

Gene expression of smokers and nonsmokers
gene: KCNE4 | 1552507_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4883
p=0.4139
p=0.5998
p=0.4979
p=0.2021
p=0.6526
p=0.5822
p=0.4625
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

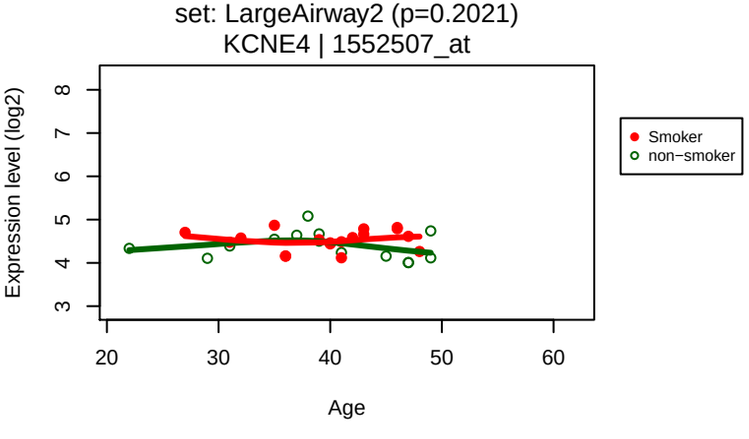

Gene expression ratio (smokers/nonsmokers)
gene: KCNE4 | 1552507_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4883
p=0.4139
p=0.5998
p=0.4979
p=0.2021
p=0.6526
p=0.5822
p=0.4625
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

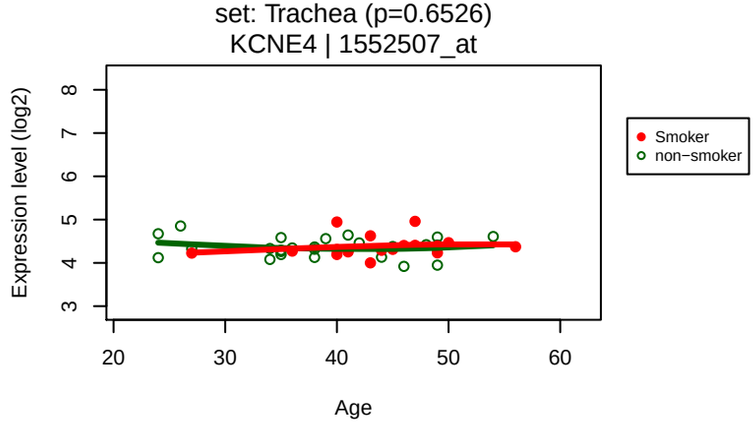

Gene expression ratio in Males
gene: KCNE4 | 1552507_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3736
p=0.9227
p=0.6755
p=0.4903
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KCNE4 | 1552507_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9741
p=0.01
p=0.7939
p=0.7685
SmallAirway2
LargeAirway2
Trachea
Macrophage2

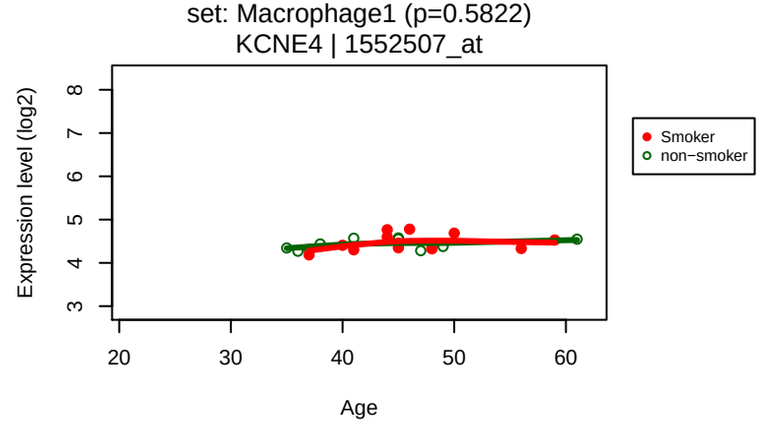

set: SmallAirway1 (p=0.4883)
KCNE4 | 1552507_at
Expression level (log2)
Age
Smoker
non−smoker

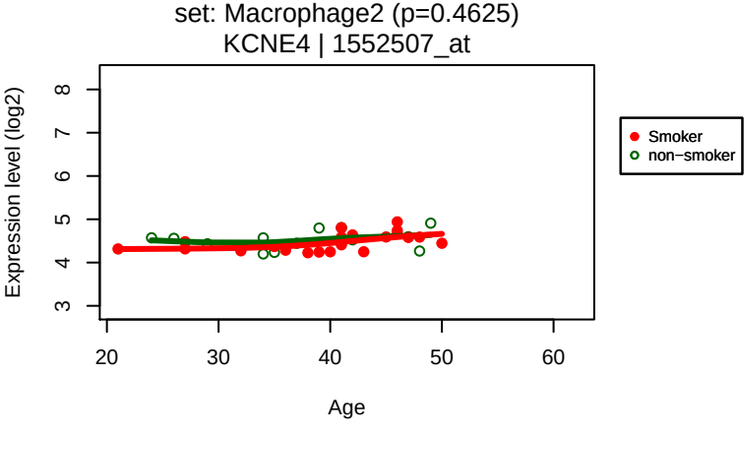

set: SmallAirway2 (p=0.4139)
KCNE4 | 1552507_at
Expression level (log2)
Age
Smoker
non−smoker

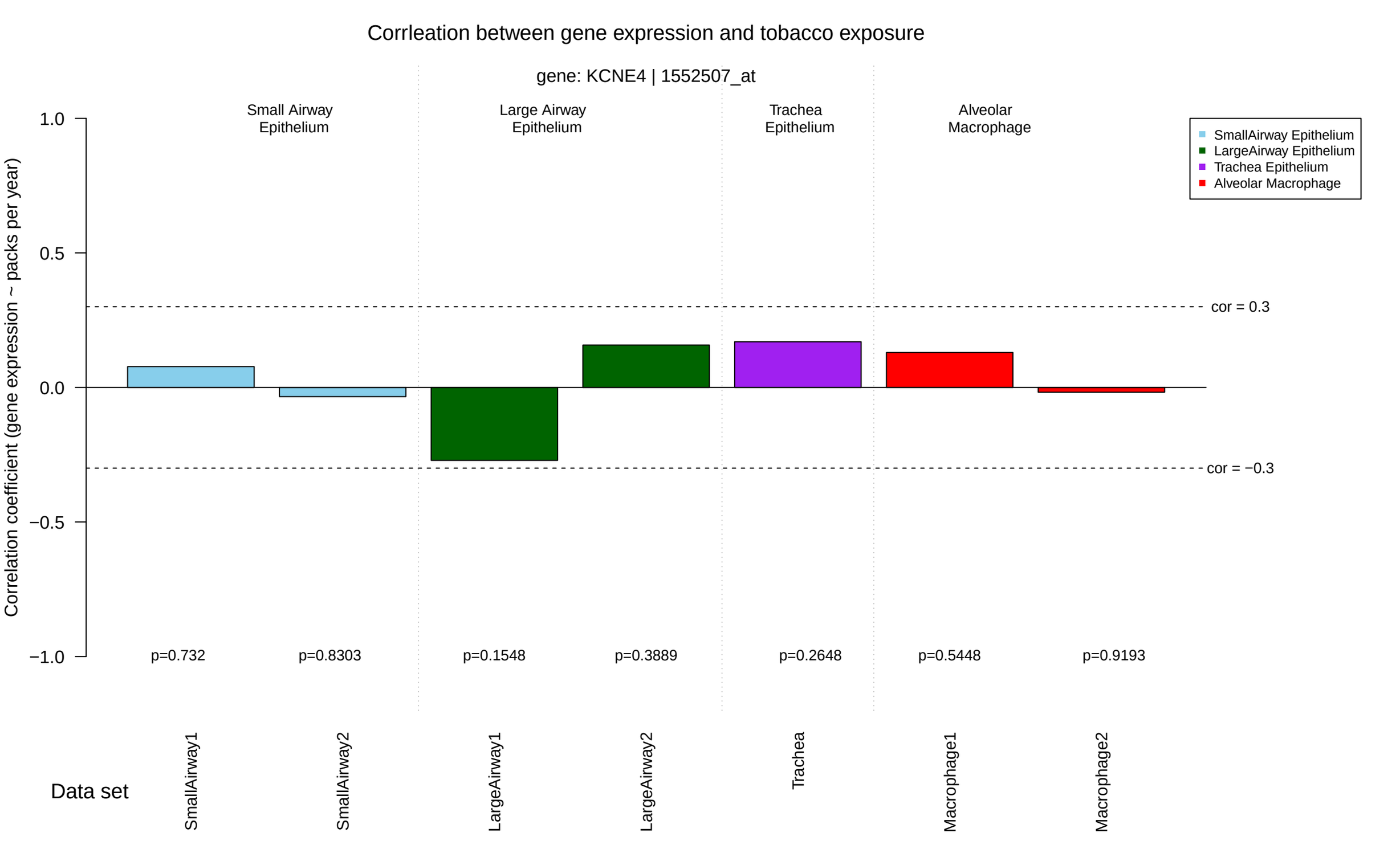

set: LargeAirway1 (p=0.4979)
KCNE4 | 1552507_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2021)
KCNE4 | 1552507_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6526)
KCNE4 | 1552507_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5822)
KCNE4 | 1552507_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4625)
KCNE4 | 1552507_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KCNE4 | 1552507_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.732
p=0.8303
p=0.1548
p=0.3889
p=0.2648
p=0.5448
p=0.9193
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2